

7. **Image 7.1** and **Image 7.2** show examples of two methods of farming hens for egg production.

**Image 7.1 – Intensively farmed chickens**



**Image 7.2 – Free-range chickens**



- Intensive farming methods maximise production by controlling the conditions in which farm animals are kept. Inside animal sheds, temperatures may be kept high and each animal is given limited space.
- In 2015, a survey of 2000 people (*OnePoll*) found that 80% of those questioned always or often bought free-range eggs, even though they were more expensive.

**Table 7.3** shows egg production in the UK by intensive and free-range methods between 2006 and 2016.

**Table 7.3**

| Year | Egg production (billion) |            |
|------|--------------------------|------------|
|      | intensive                | free-range |
| 2006 | 4.1                      | 1.9        |
| 2008 | 4.0                      | 2.1        |
| 2010 | 3.8                      | 2.8        |
| 2012 | 3.5                      | 3.4        |
| 2014 | 3.9                      | 3.3        |
| 2016 | 3.8                      | 3.8        |

- (a) Use **Table 7.3** and the information above to answer the following questions.
- (i) Calculate the percentage increase in the production of **free-range eggs** between 2006 and 2016. [2]

Increase = ..... %



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(ii) Give **one** way that intensive farming methods minimise energy loss from farm animals. Explain your answer. [2]

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(b) Suggest **two** reasons why many people buy free-range eggs, even though they are more expensive than eggs produced by intensive farming methods. [2]

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(c) (i) State **one** possible **pollutant** from intensive farming methods and explain how it could damage the environment. [2]

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(ii) Farmers in Wales who plan to develop intensive methods of food production on rural land must first submit their plans to biologists at Natural Resources Wales. Describe the role of the biologists at both the planning stage and when the intensive farm is operating fully. [2]

Planning stage

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When operating fully

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